

Models: Fitting and Evaluation

Table of contents

Microplastics	1
Total model	1
Model evaluation	3
Bottled model	5
Model evaluation	8
Fisher exact tests	11
PE model	13
Model evaluation	14
PET model	18
Model evaluation	20
PFAS	24
Total model	24
Model evaluation	26
Bottled model	28
Fisher exact tests	29

Microplastics

Total model

```
Family: nbinom2 ( log )
Formula:
particles ~ category + bottle_material + source_type + adv_filtration
Zero inflation:      ~.
Data: mp.total
```

AIC	BIC	logLik	-2*log(L)	df.resid
-----	-----	--------	-----------	----------

414.6 435.9 -196.3 392.6 40

Dispersion parameter for nbinom2 family (): 1.22

Conditional model:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	3.1512	0.5717	5.512	3.55e-08	***
categoryBottled	1.9115	0.6683	2.860	0.00423	**
bottle_materialPET	-1.5790	0.3624	-4.358	1.32e-05	***
bottle_materialHDPE	NA	NA	NA	NA	
source_typeSpring	-0.2986	0.4329	-0.690	0.49037	
adv_filtrationYes	0.2620	0.4951	0.529	0.59667	

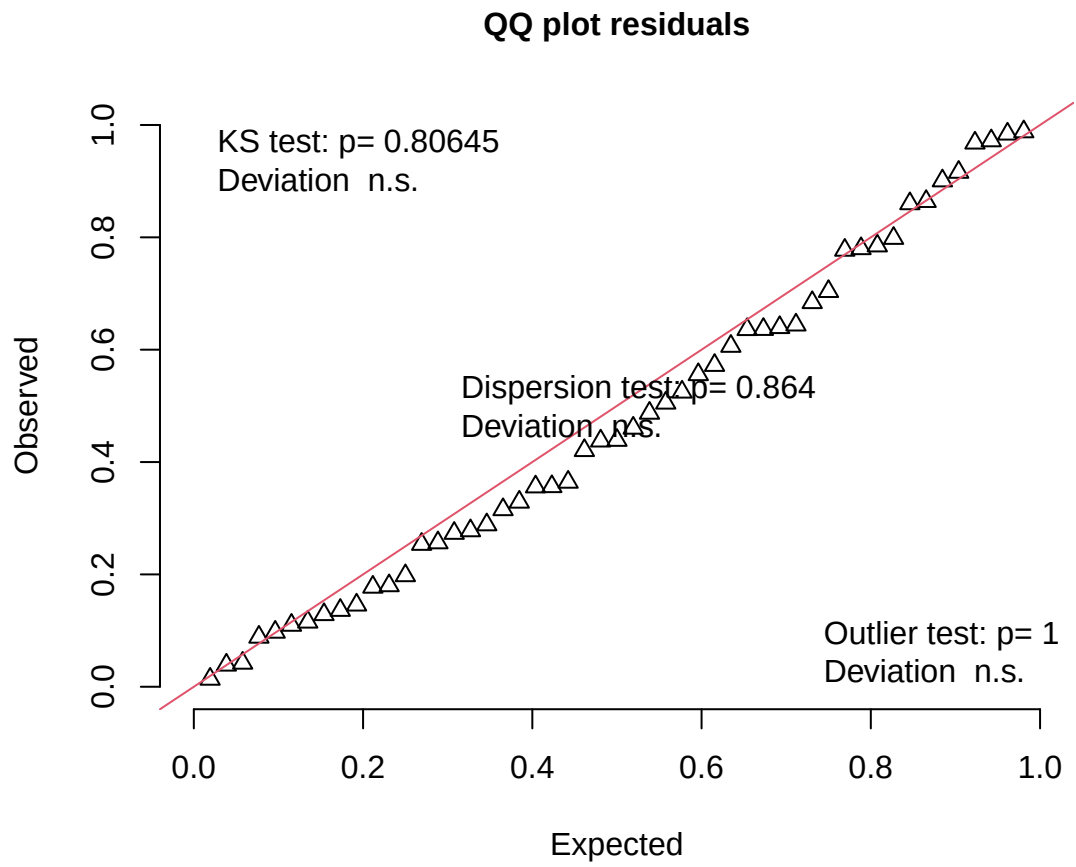
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

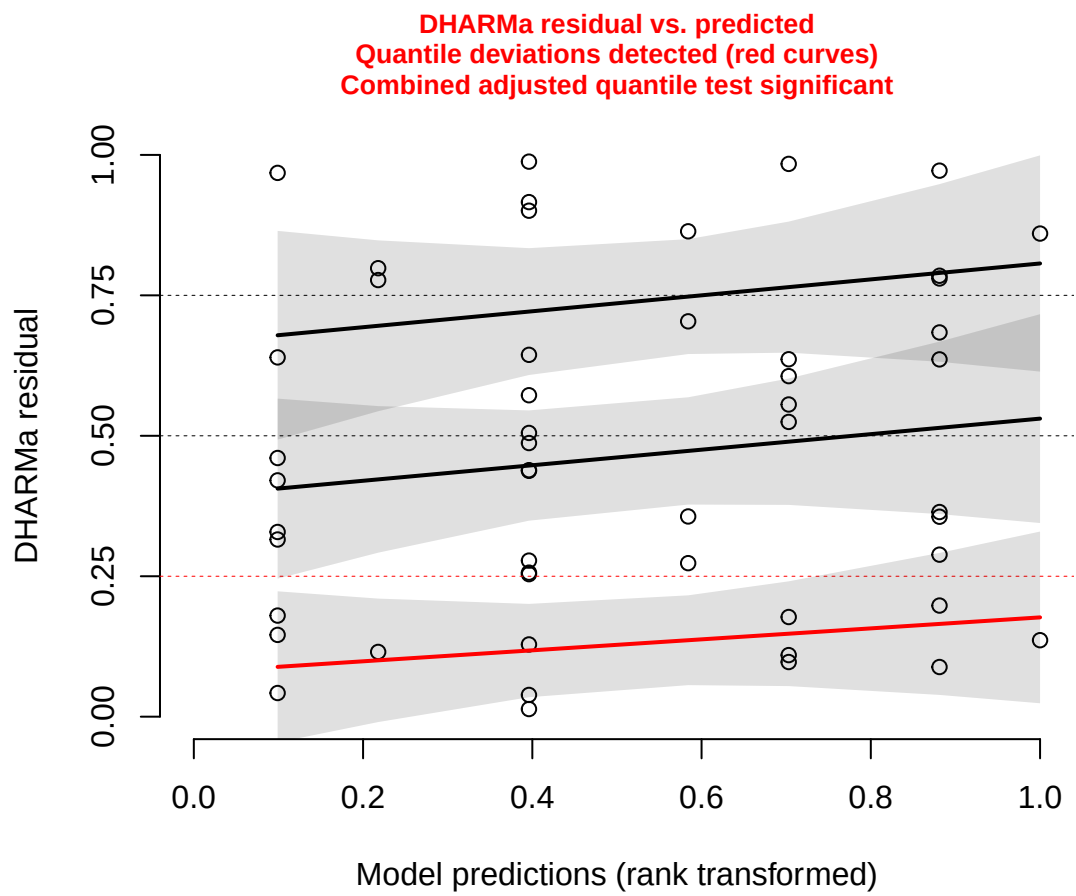
Zero-inflation model:

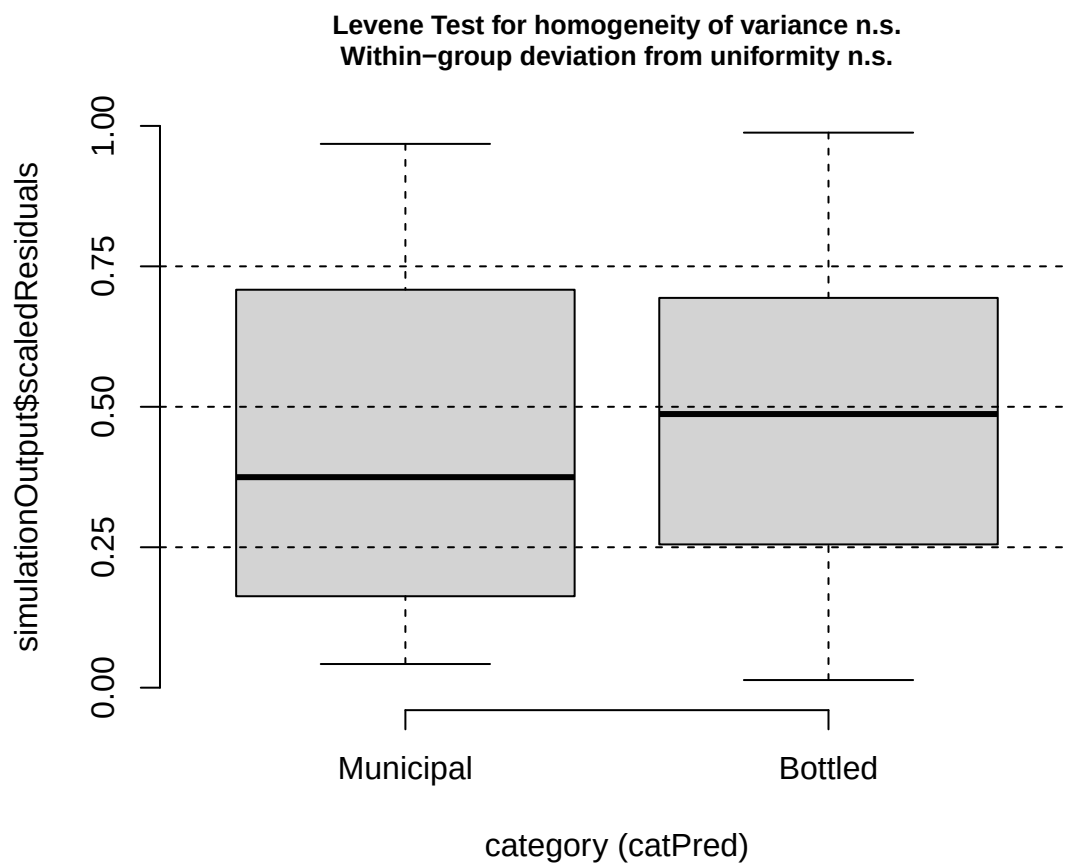
	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	1.2563	0.7636	1.645	0.0999	.
categoryBottled	-3.6909	1.5543	-2.375	0.0176	*
bottle_materialPET	1.0085	1.2452	0.810	0.4180	
bottle_materialHDPE	NA	NA	NA	NA	
source_typeSpring	-0.0125	1.1606	-0.011	0.9914	
adv_filtrationYes	-0.6730	1.1230	-0.599	0.5490	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Model evaluation







Bottled model

```
Family: nbinom2 ( log )
Formula:          particles ~ bottle_material + source_type + adv_filtration
Zero inflation:    ~.
Data: mp.bottled.total
```

AIC	BIC	logLik	-2*log(L)	df.resid
369.5	384.5	-175.8	351.5	30

Dispersion parameter for nbinom2 family (): 1.19

Conditional model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	3.3120	0.4030	8.218	< 2e-16 ***
bottle_materialHDPE	1.6313	0.3646	4.474	7.67e-06 ***
source_typeSpring	-0.1514	0.4318	-0.351	0.726
adv_filtrationYes	0.5816	0.5534	1.051	0.293

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Zero-inflation model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-19.3246	7037.5151	-0.003	0.998
bottle_materialHDPE	-0.7074	1.3152	-0.538	0.591
source_typeSpring	17.7934	7037.5151	0.002	0.998
adv_filtrationYes	18.1784	7037.5152	0.003	0.998

The large standard errors in the zero-inflated component are because `source_type` and `adv_filtration` are very collinear. We can see in the below table that all samples with “Yes” for advanced filtration come from non-spring sources.

```
table(mp.bottled.total$source_type, mp.bottled.total$adv_filtration)
```

	No	Yes
Non-spring	6	8
Spring	25	0

So, we do not look at the zero-inflated portion of the model, as the estimates and inference are not reliable.

We check that neither variable is significant on its own:

```
Family: nbinom2 ( log )
Formula:          particles ~ bottle_material + source_type
Zero inflation:    ~.
Data: mp.bottled.total
```

AIC	BIC	logLik	-2*log(L)	df.resid
368.6	380.2	-177.3	354.6	32

Dispersion parameter for nbinom2 family (): 1.13

Conditional model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	3.6468	0.2814	12.959	< 2e-16 ***
bottle_materialHDPE	1.5392	0.3702	4.158	3.21e-05 ***
source_typeSpring	-0.4477	0.3641	-1.229	0.219

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Zero-inflation model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.8168	0.8669	-2.096	0.0361 *
bottle_materialHDPE	-0.9295	1.2716	-0.731	0.4648
source_typeSpring	0.3336	1.0769	0.310	0.7567

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Family: nbinom2 (log)

Formula: particles ~ bottle_material + adv_filtration

Zero inflation: ~.

Data: mp.bottled.total

AIC	BIC	logLik	-2*log(L)	df.resid
367.4	379.0	-176.7	353.4	32

Dispersion parameter for nbinom2 family (): 1.16

Conditional model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	3.1908	0.2386	13.371	< 2e-16 ***
bottle_materialHDPE	1.6317	0.3685	4.428	9.5e-06 ***
adv_filtrationYes	0.7018	0.4513	1.555	0.12

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

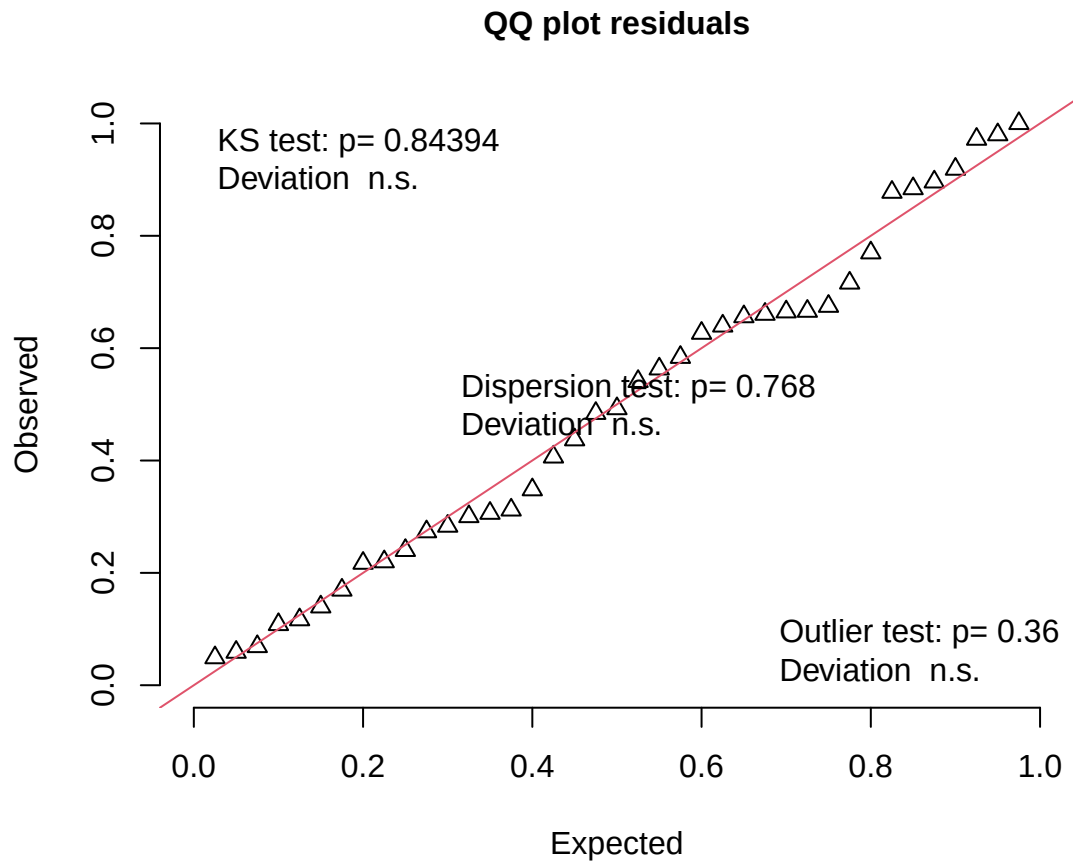
Zero-inflation model:

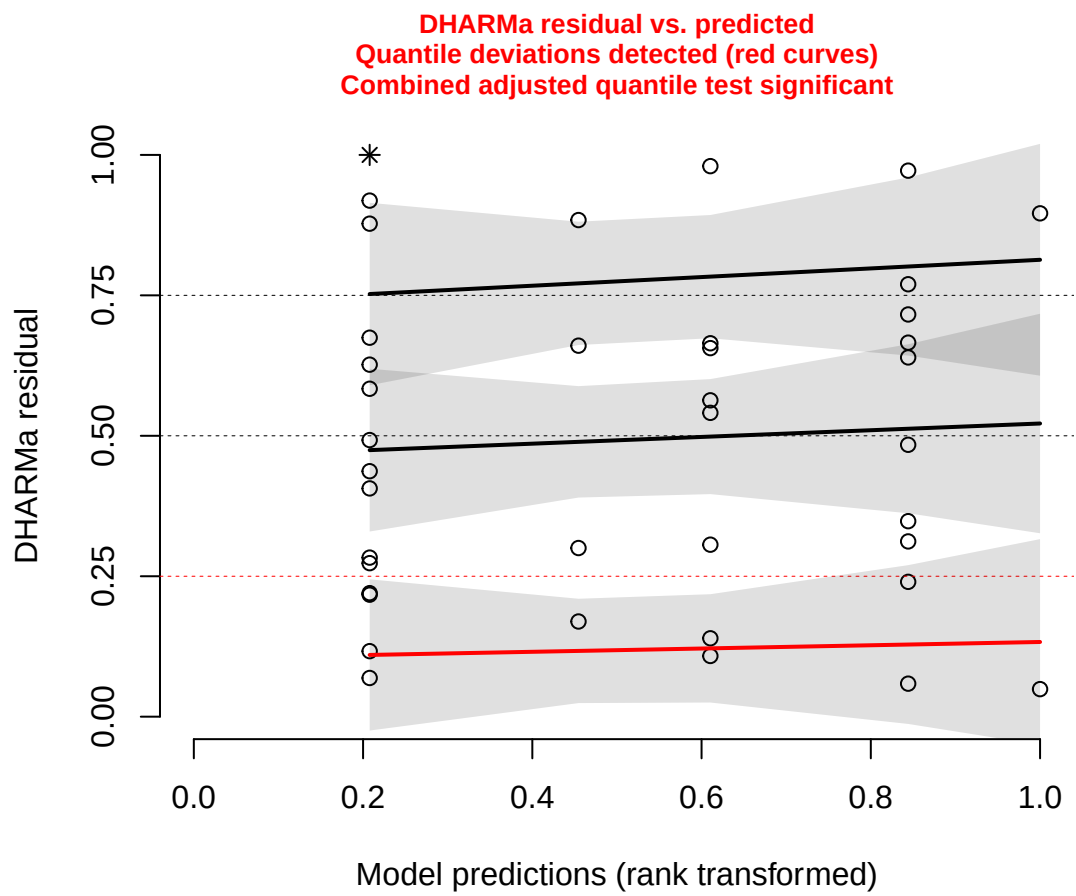
	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.8664	0.7579	-2.463	0.0138 *
bottle_materialHDPE	-0.5849	1.3280	-0.440	0.6596
adv_filtrationYes	0.7163	1.1332	0.632	0.5273

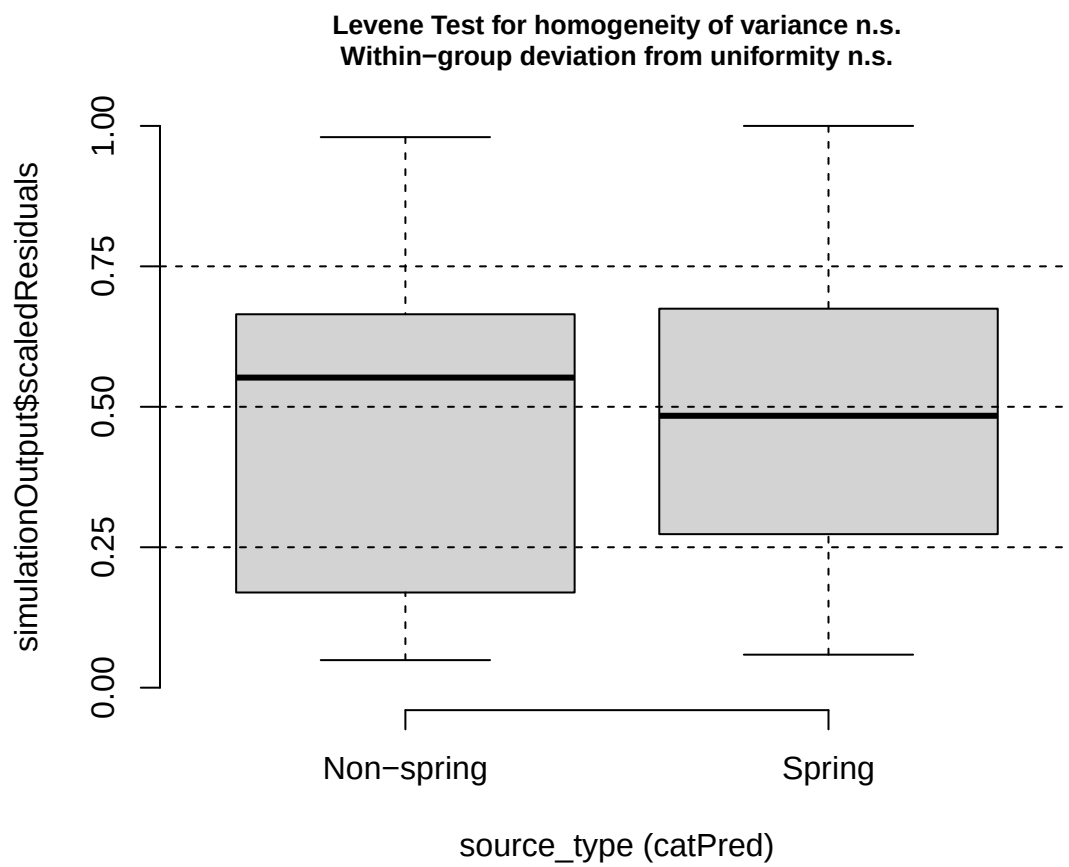
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

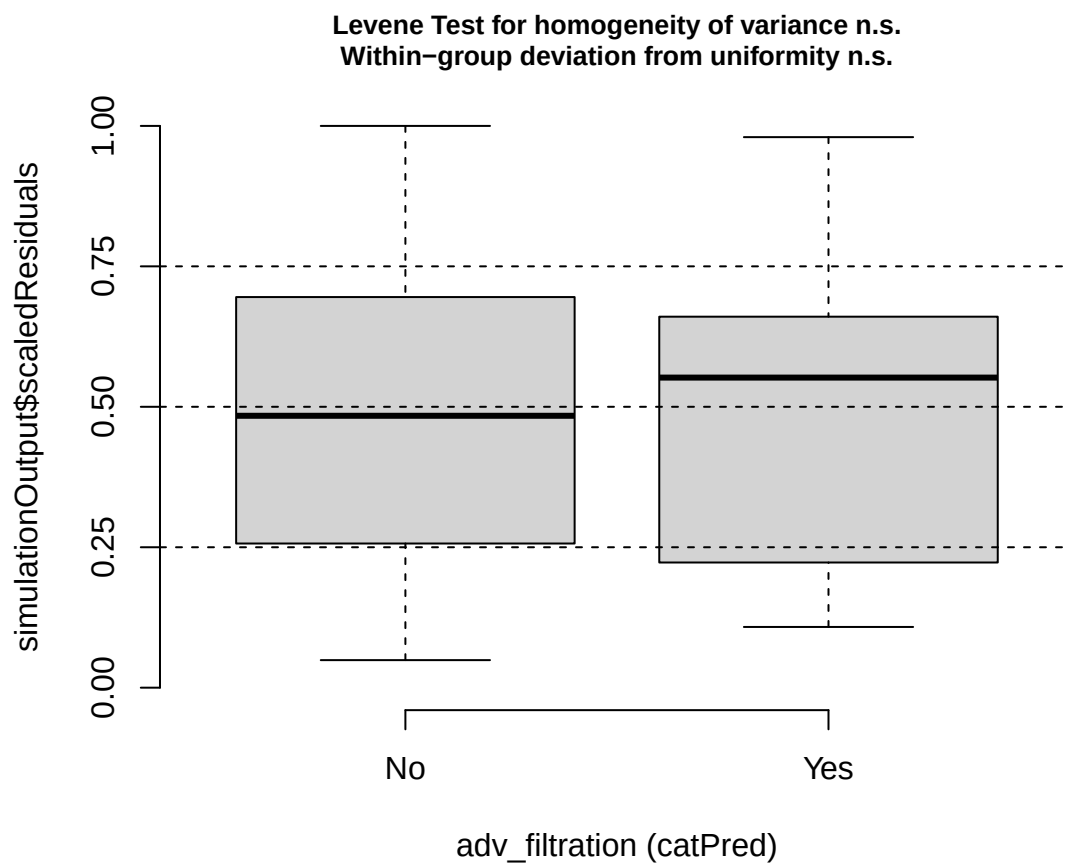
Because neither variable is significant in the zero-inflated component on its own, we are reasonably confident there is no difference in detection probability by either source type or advanced filtration. To report p-values, we instead report the Fisher exact test p-values, which don't control for other variables, as the p-values from the zero-inflated portion of the model are not reliable. See below for the Fisher exact tests.

Model evaluation









Fisher exact tests

Probability of detection: source

[1] 0.84

[1] 0.8571429

	0	1
Non-spring	2	12
Spring	4	21

Exact Two-Sample Fisher-Pitman Permutation Test

data: detect by factor(source_type) (Non-spring, Spring)
Z = 0.1405, p-value = 1
alternative hypothesis: true mu is not equal to 0

Probability of detection: bottle material

	0	1
PET	5	22
HDPE	1	11

Exact Two-Sample Fisher-Pitman Permutation Test

data: detect by factor(bottle_material) (PET, HDPE)
Z = -0.80316, p-value = 0.645
alternative hypothesis: true mu is not equal to 0

Probability of detection: advanced filtration

[1] 0.75

[1] 0.8709677

	Non-spring	Spring
No	6	25
Yes	8	0

	0	1
No	4	27
Yes	2	6

Exact Two-Sample Fisher-Pitman Permutation Test

data: detect by factor(adv_filtration) (No, Yes)
Z = 0.83455, p-value = 0.5834
alternative hypothesis: true mu is not equal to 0

PE model

Family: nbinom2 (log)
Formula: particles ~ bottle_material + source_type + adv_filtration
Zero inflation: ~.
Data: mp.bottled.pe

AIC	BIC	logLik	-2*log(L)	df.resid
325.6	340.6	-153.8	307.6	30

Dispersion parameter for nbinom2 family (): 0.897

Conditional model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	4.0518	0.4965	8.161	3.34e-16 ***
bottle_materialPET	-1.4517	0.4692	-3.094	0.00198 **
source_typeSpring	0.5429	0.5590	0.971	0.33144
adv_filtrationYes	1.1332	0.7293	1.554	0.12026

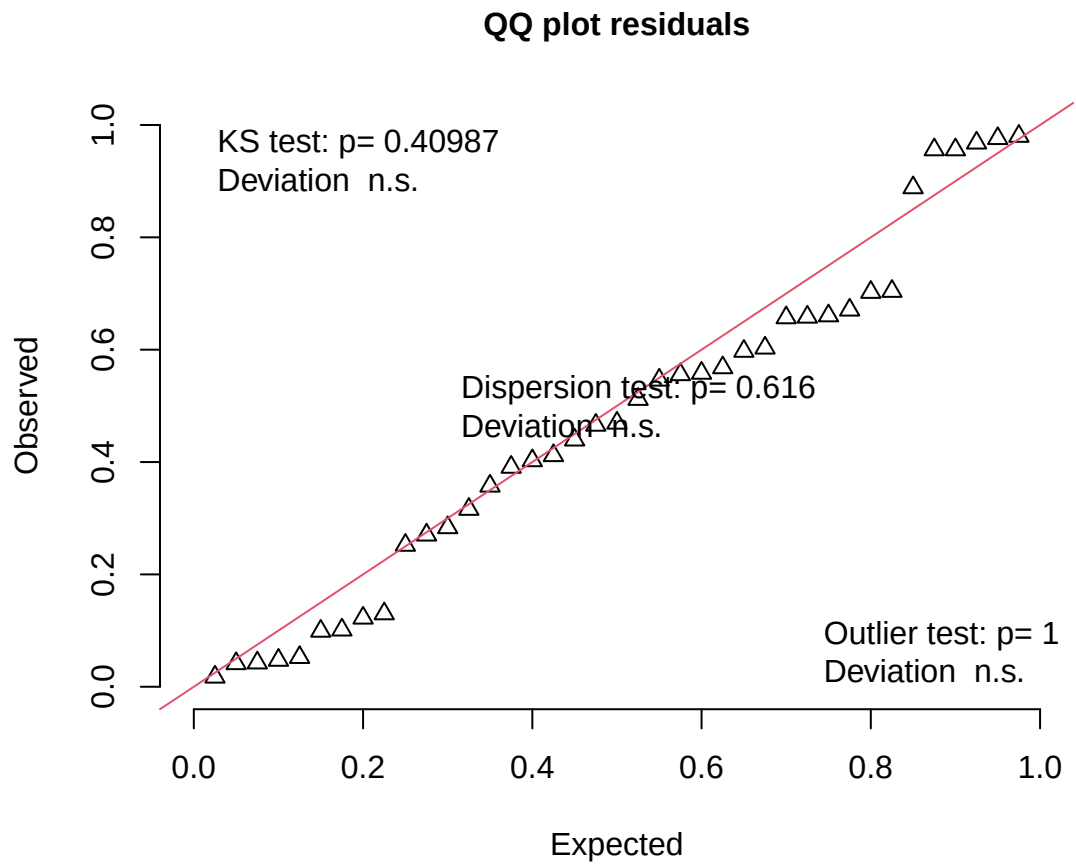
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

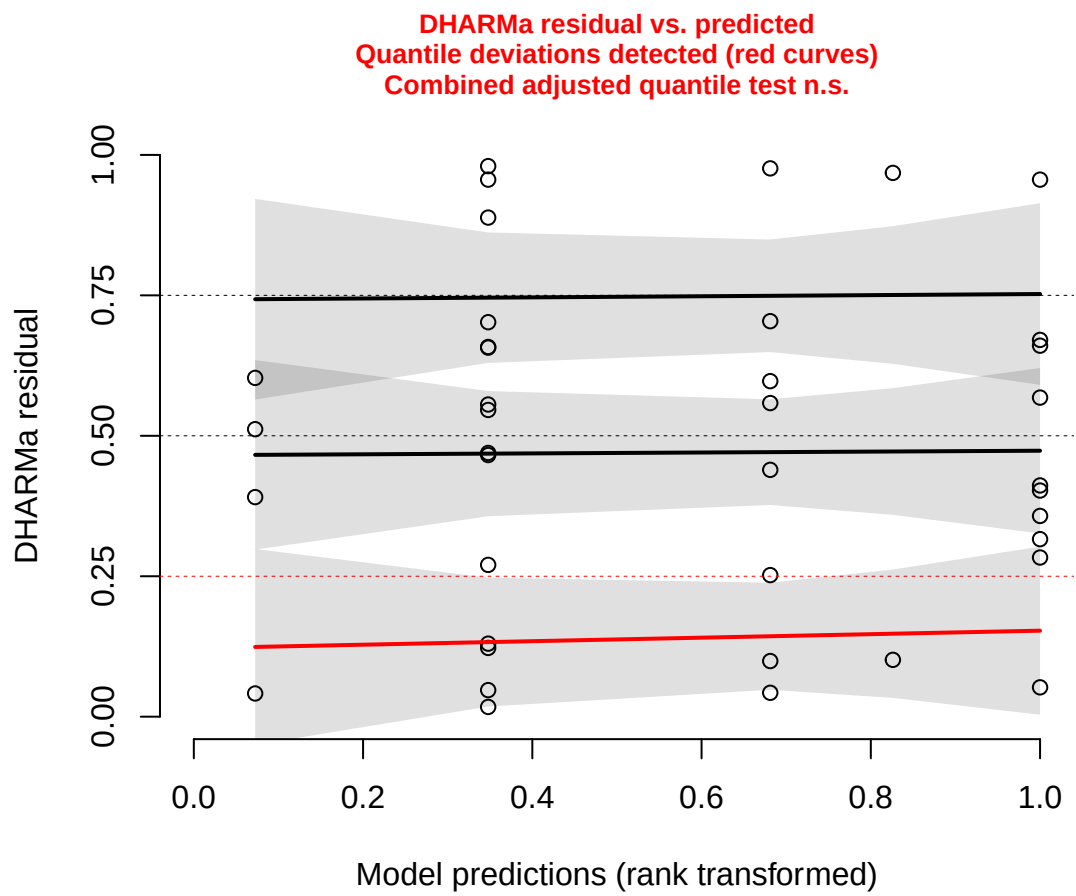
Zero-inflation model:

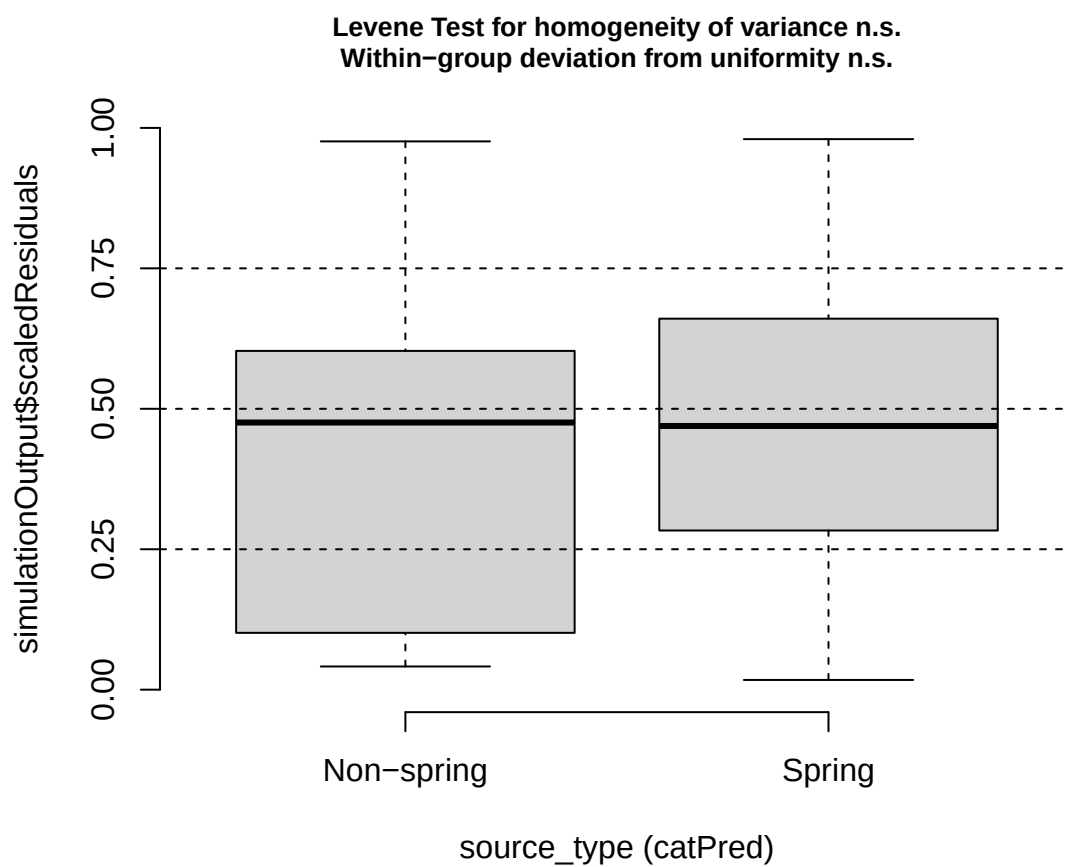
	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-3.8033	2.0454	-1.859	0.063 .
bottle_materialPET	2.1828	1.3356	1.634	0.102
source_typeSpring	1.3814	1.7203	0.803	0.422
adv_filtrationYes	0.3883	1.8810	0.206	0.836

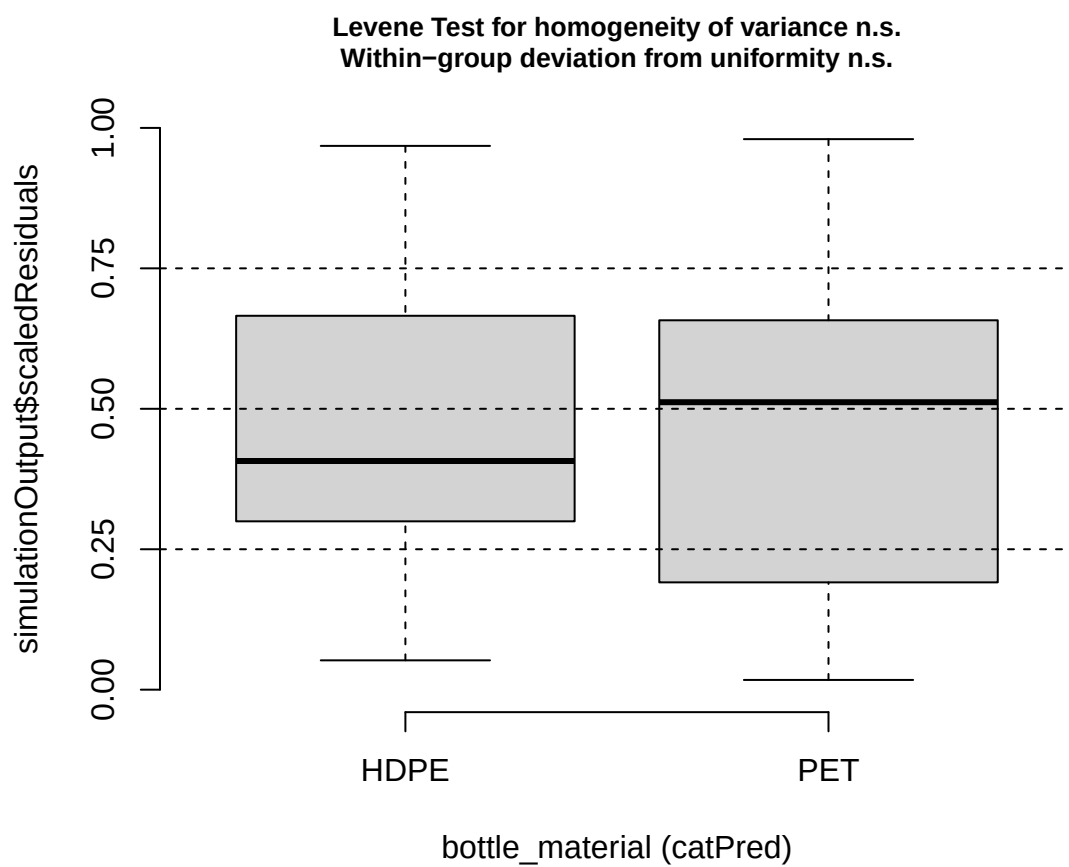
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

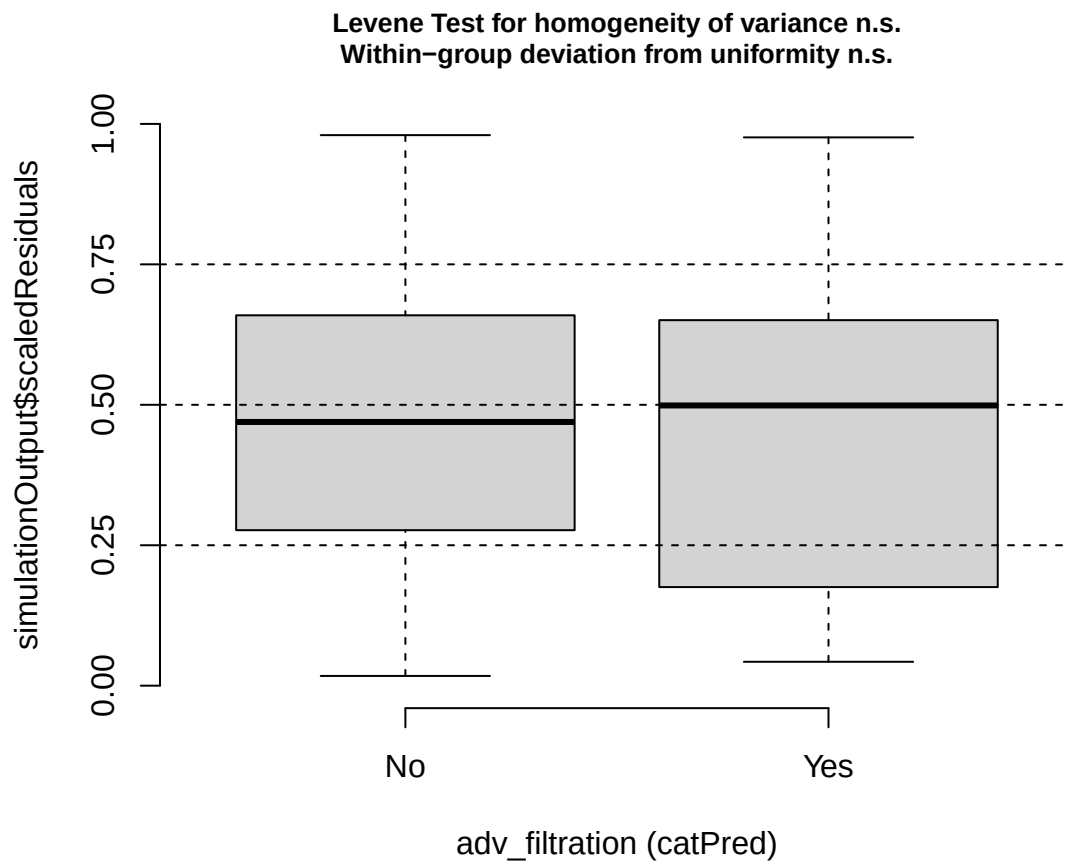
Model evaluation











PET model

```
Family: nbinom2 ( log )
Formula:          particles ~ source_type + bottle_material + adv_filtration
Zero inflation:    ~.
Data: mp.bottled.pet
```

AIC	BIC	logLik	-2*log(L)	df.resid
174.4	189.3	-78.2	156.4	30

Dispersion parameter for nbinom2 family (): 3.49

Conditional model:

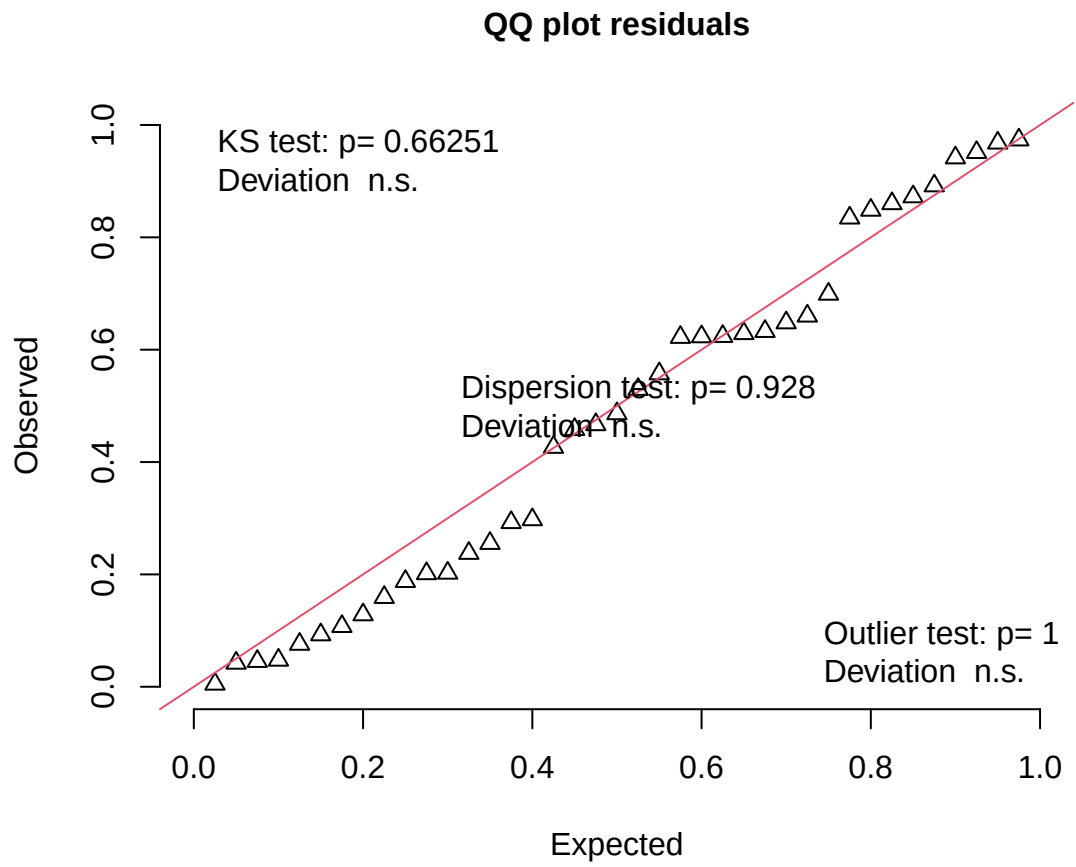
	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	3.2052	0.4468	7.174	7.3e-13 ***
source_typeSpring	0.1953	0.4714	0.414	0.67871
bottle_materialPET	-1.0995	0.3512	-3.131	0.00174 **
adv_filtrationYes	0.4547	0.5932	0.767	0.44333

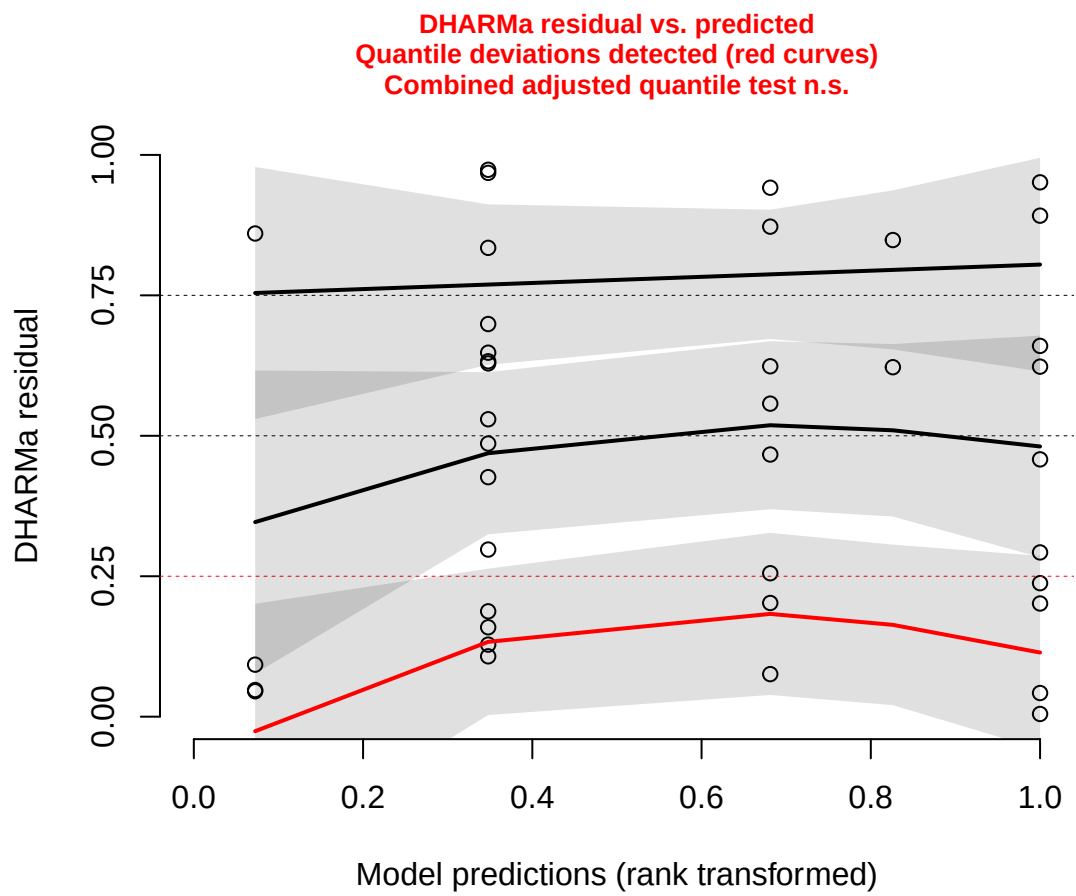
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

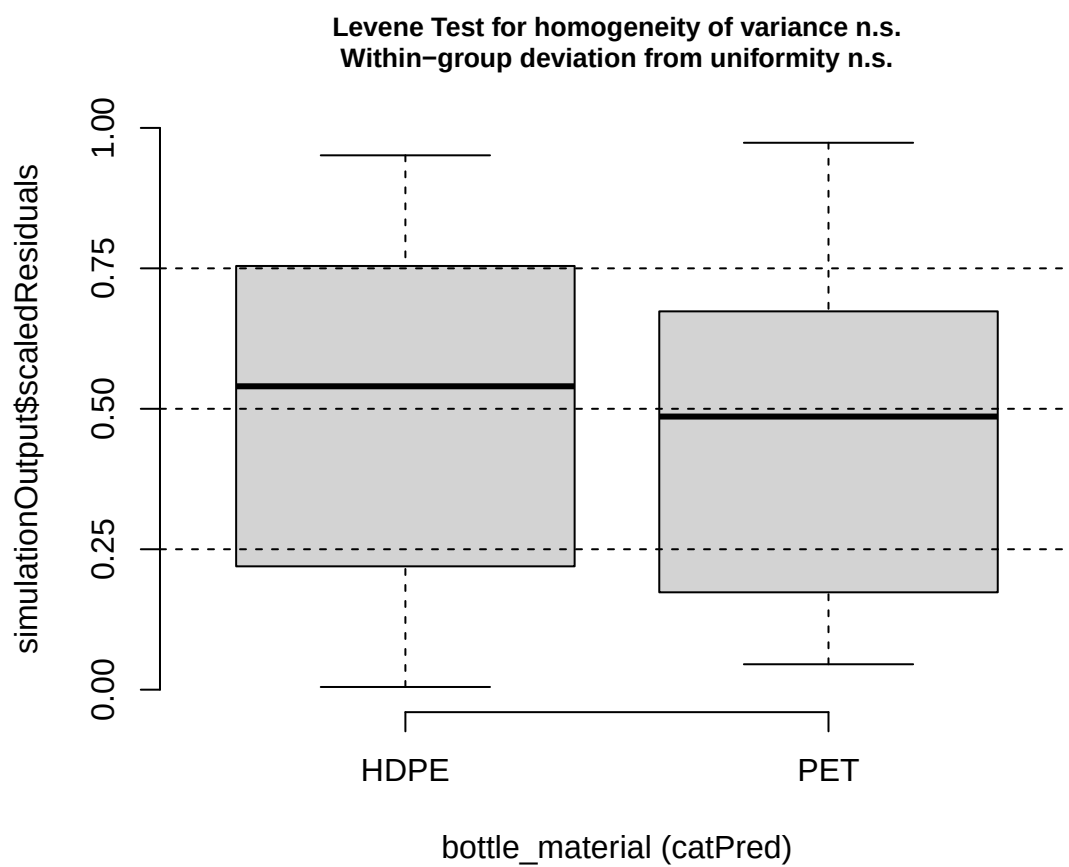
Zero-inflation model:

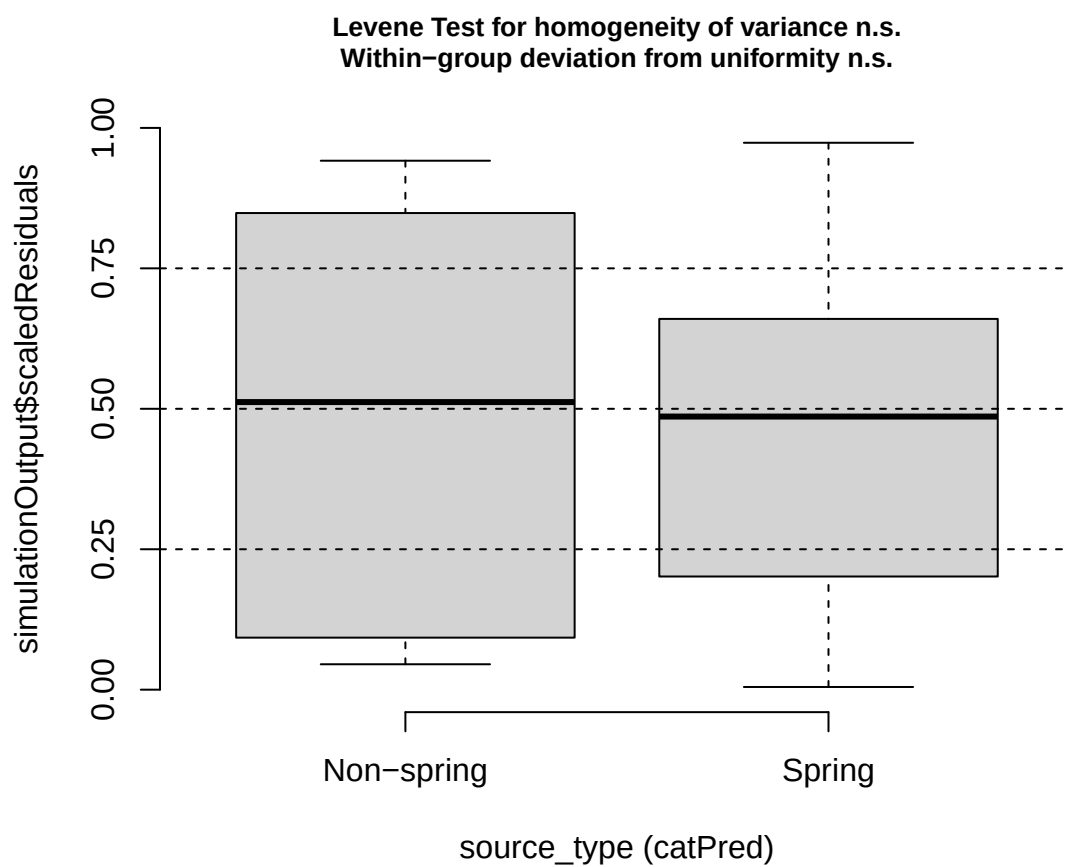
	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.5609	1.0008	0.560	0.575
source_typeSpring	-0.2691	0.9643	-0.279	0.780
bottle_materialPET	0.1761	0.7583	0.232	0.816
adv_filtrationYes	-0.2334	1.1691	-0.200	0.842

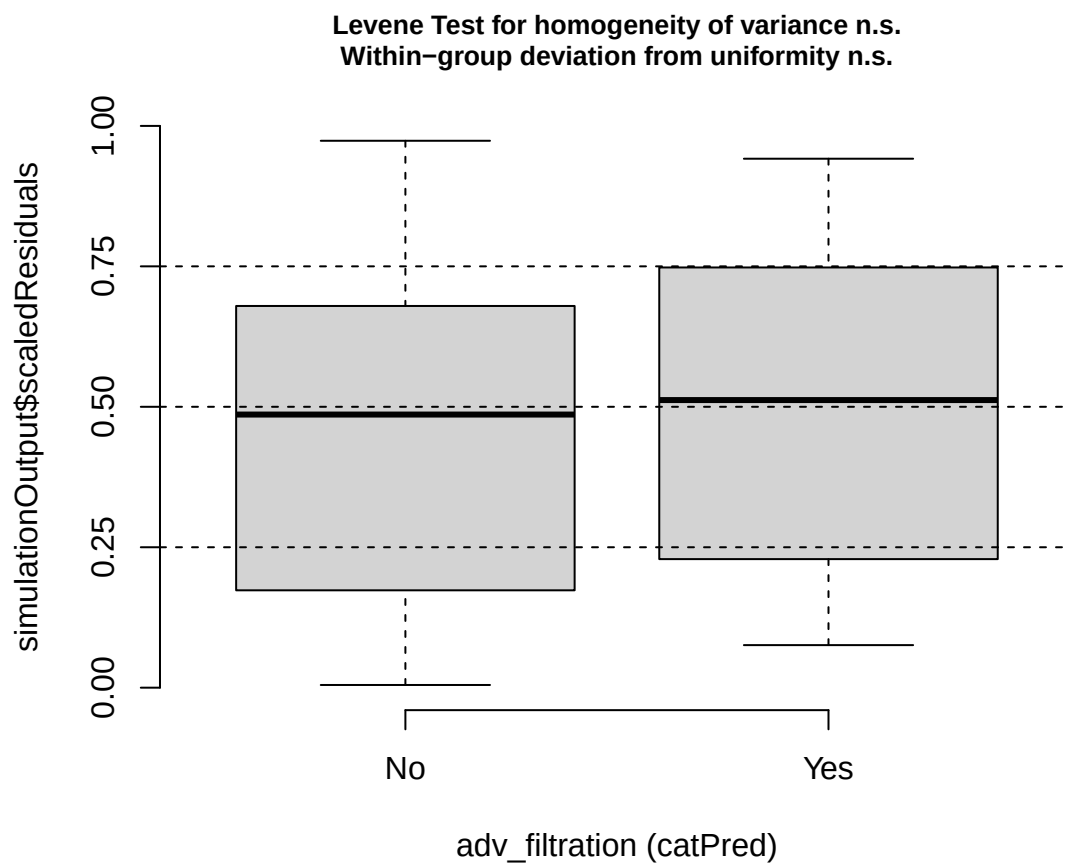
Model evaluation











PFAS

Total model

```
Family: Gamma ( log )
Formula:      concentration ~ category
Zero inflation: ~.
Data: pfas.total
```

AIC	BIC	logLik	-2*log(L)	df.resid
523.7	533.5	-256.9	513.7	47

Dispersion estimate for Gamma family (σ^2): 0.368

Conditional model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	5.5472	0.1264	43.88	< 2e-16 ***
categoryMunicipal	0.9889	0.2223	4.45	8.61e-06 ***

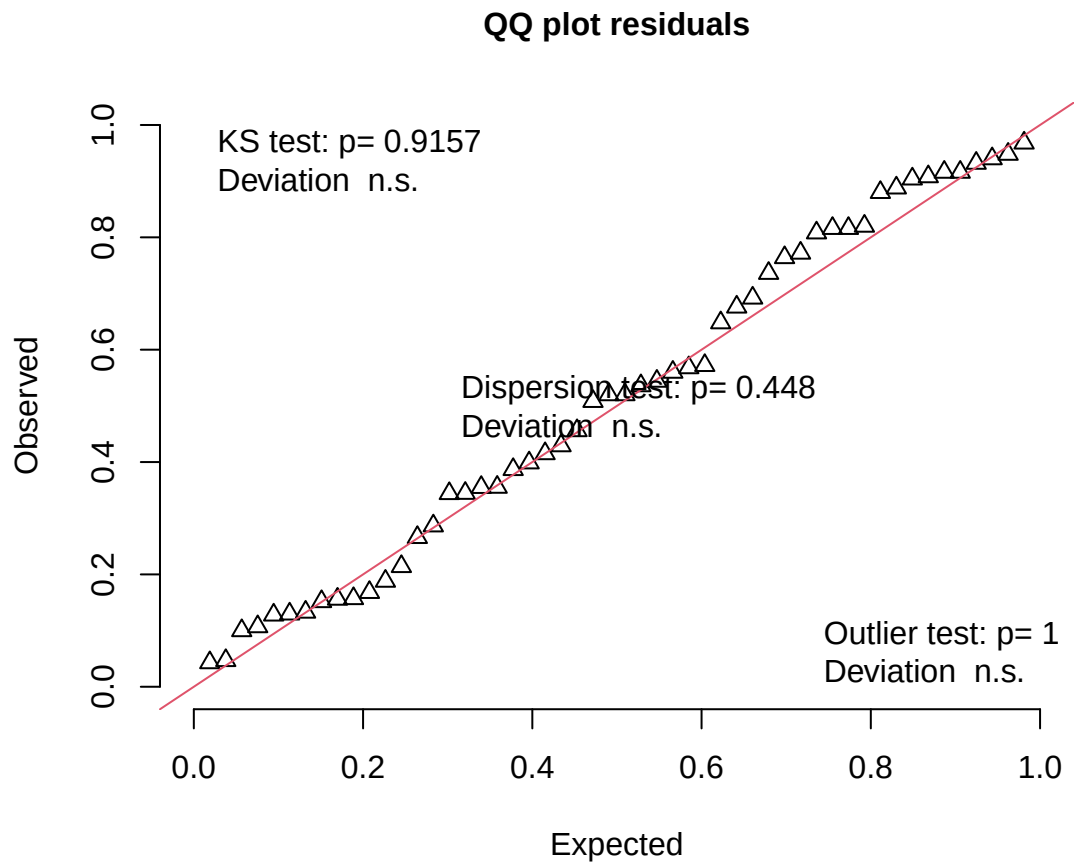
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

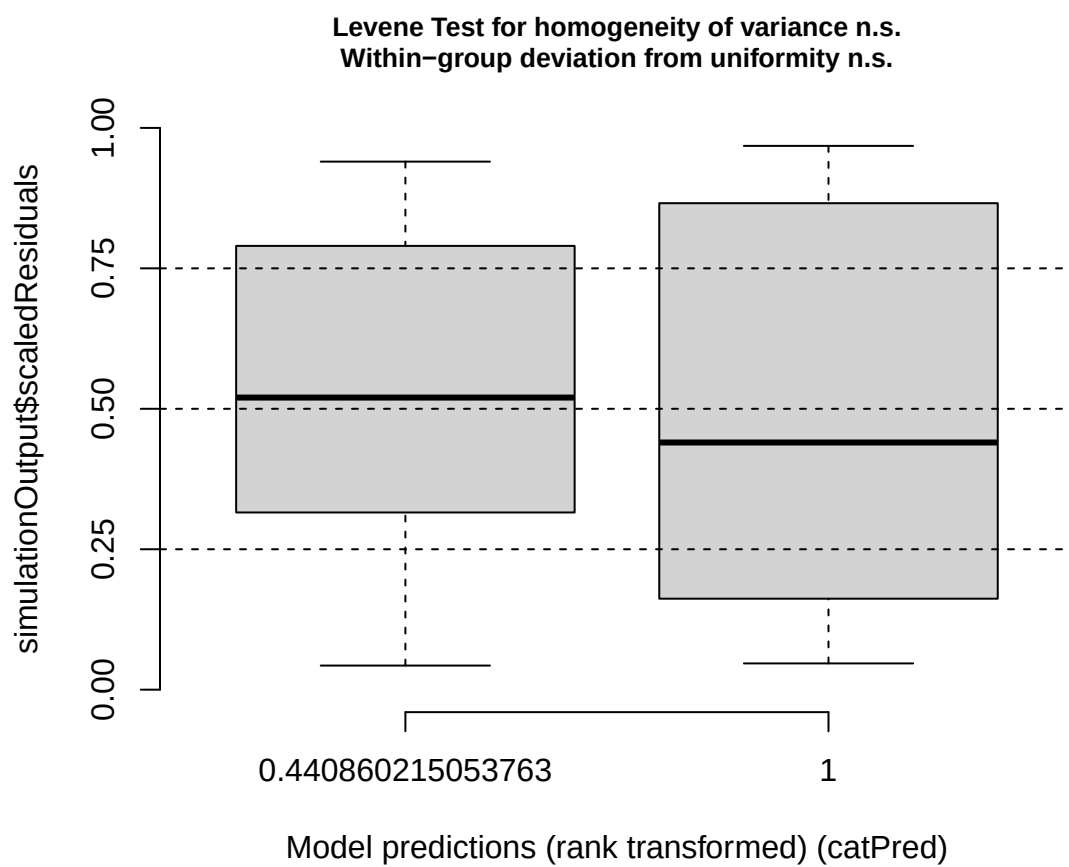
Zero-inflation model:

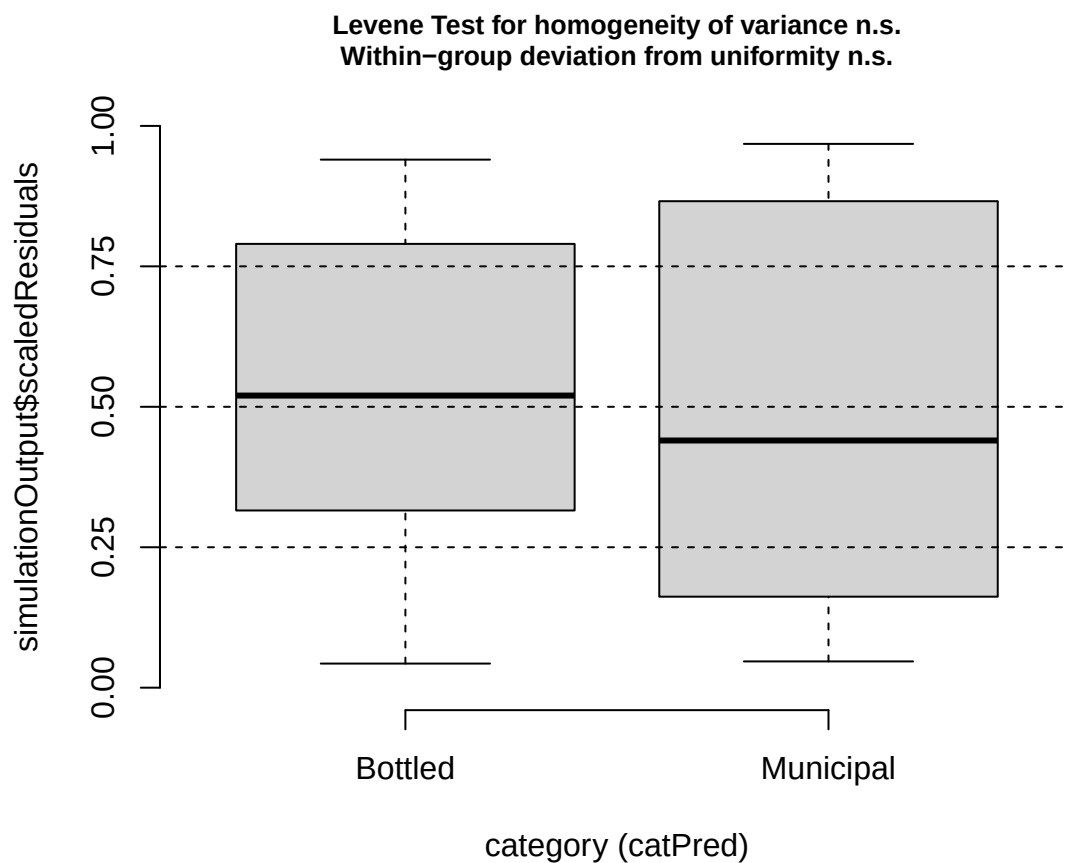
	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.3023	0.3198	-0.945	0.3446
categoryMunicipal	-2.0956	1.0923	-1.919	0.0551 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Model evaluation







Bottled model

```
Family: Gamma ( log )
Formula:      concentration ~ source_type + adv_filtration
Zero inflation: ~.
Data: pfas.bottled.total
```

AIC	BIC	logLik	-2*log(L)	df.resid
NA	NA	NA	NA	33

Dispersion estimate for Gamma family (σ^2): 0.268

Conditional model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	2.774	NaN	NaN	NaN
source_typeSpring	2.774	NaN	NaN	NaN
adv_filtrationYes	0.000	NaN	NaN	NaN

Zero-inflation model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	19.800	NaN	NaN	NaN
source_typeSpring	-21.837	NaN	NaN	NaN
adv_filtrationYes	4.067	NaN	NaN	NaN

The model does not converge due to the collinearity of source type and advanced filtration. To report p-values, we instead report the Fisher exact test p-values, which don't control for other variables, as the p-values from the zero-inflated portion of the model are not reliable. See below for the Fisher exact tests.

Fisher exact tests

Probability of detection: source

```
# A tibble: 2 x 5
  source_type num.detect num.zero total percent
<fct>         <int>      <int> <int>    <dbl>
1 Non-spring      0        14    14      0
2 Spring          23         3    26    88.5
```

Exact Two-Sample Fisher-Pitman Permutation Test

```
data: detect by factor(source_type) (Non-spring, Spring)
Z = -5.3303, p-value = 2.93e-08
alternative hypothesis: true mu is not equal to 0
```

Probability of detection: advanced filtration

```
# A tibble: 2 x 5
  adv_filtration num.detect num.zero total percent
<fct>           <int>      <int> <int>    <dbl>
1 No             23         9    32    71.9
2 Yes            0         8     8     0
```

	Non-spring	Spring
No	6	26
Yes	8	0

Exact Two-Sample Fisher-Pitman Permutation Test

data: detect by factor(adv_filtration) (No, Yes)
 Z = 2.9633, p-value = 0.004812
 alternative hypothesis: true mu is not equal to 0